SEMANA 6:

STEP 5 - STEP 6
"MATCHING CON STEP 1 - SELECCIÓN DE MÉTODOS DE D.M."

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9 STEPS

Fuente: Fayyad, Piatetsky-Shapiro, & Smyth (1996)

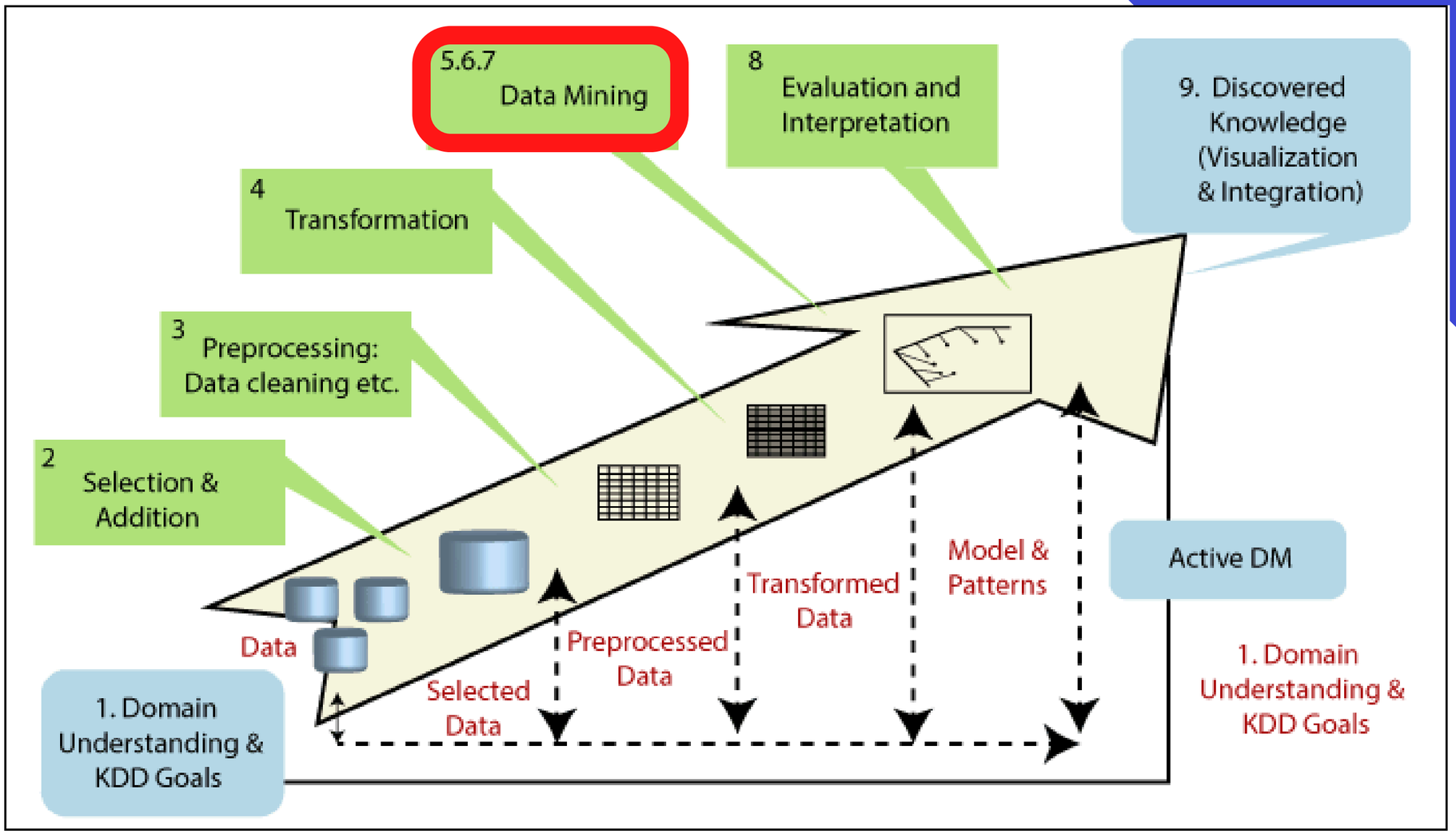


Imagen: https://www.javatpoint.com/kdd-process-in-data-mining

Comparativo de Metodologías

Data Mining Process Models	KDD	CRISP-DM	SEMMA
No. of Steps	9	6	5
Name of Steps	Developing and Understanding of the Application	Business Understanding	---------
	Creating a Target Data Set	Data Understanding	Sample
	Data Cleaning and Pre-processing		Explore
	Data Transformation	Data Preparation	Modify
	Choosing the suitable Data Mining Task	Modeling	Model
	Choosing the suitable Data Mining Algorithm		
	Employing Data Mining Algorithm		
	Interpreting Mined Patterns	Evaluation	Assessment
	Using Discovered Knowledge	Deployment	---------

Imagen: https://medium.com/analytics-vidhya/knowledge-discovery-data-kdd-a8b41509bff9

¿PREDECIR?
¿DESCRIBIR?
¿OPTIMIZAR?

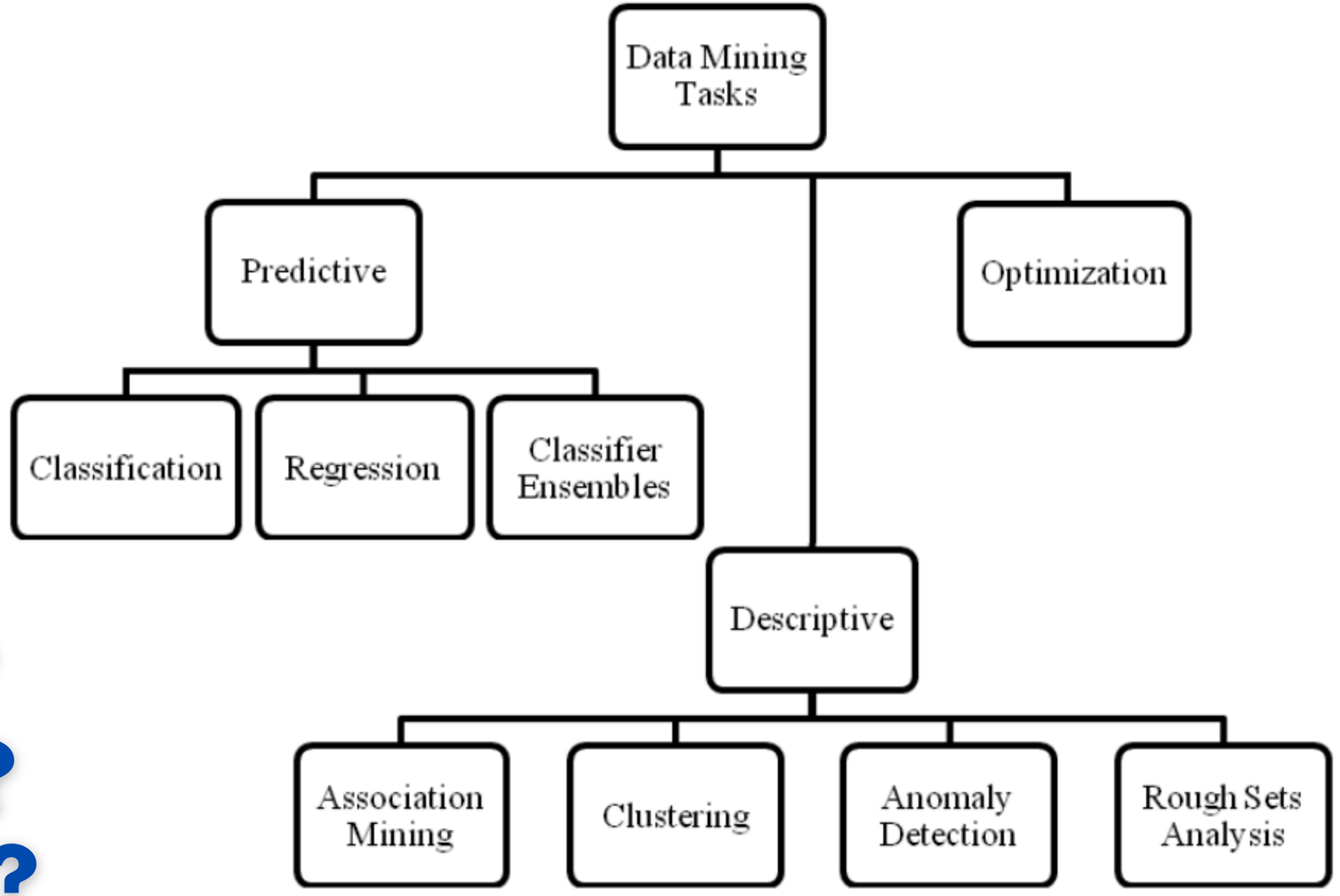
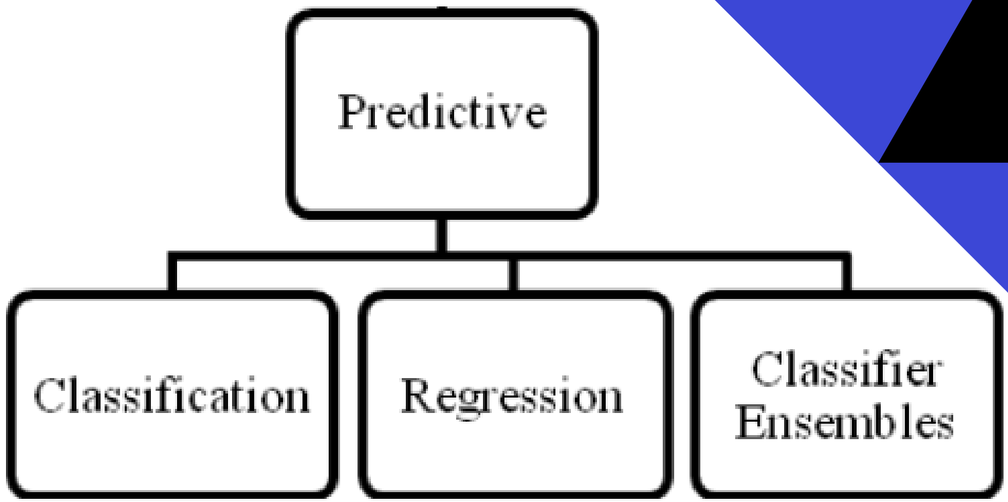


Imagen: EvolutionofDataMiningTechniquestoBigDataAnalytics.pdf

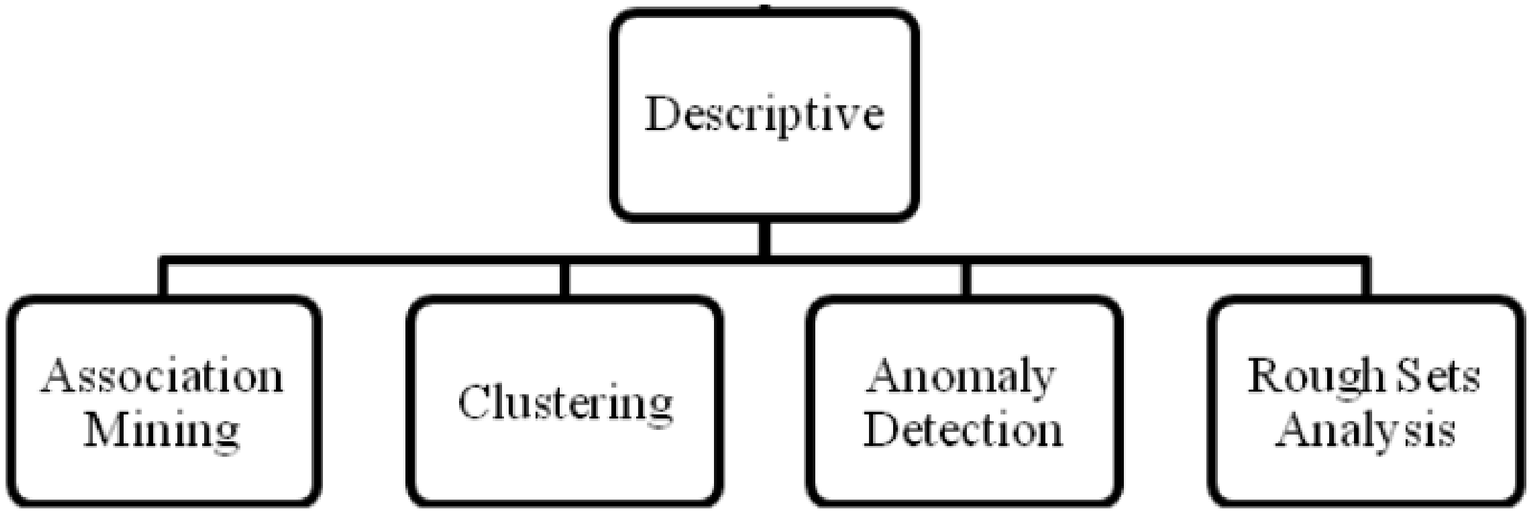
Predictive

Usa variables específicas del conjunto de datos para predecir valores desconocidos o futuros de determinada variable o variables de interés



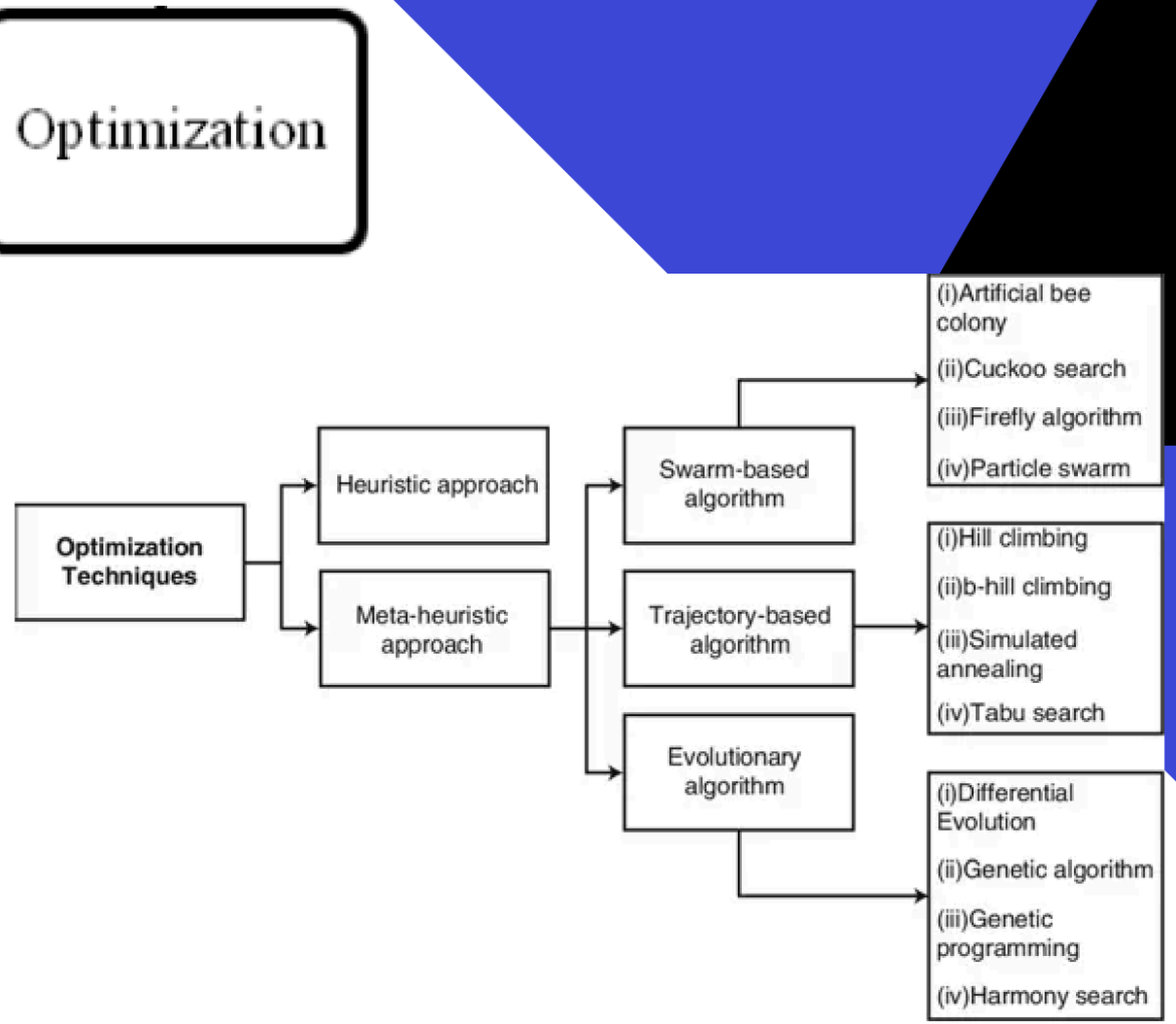
Descriptive

Cuantifica relaciones entre los datos, analizando eventos pasados en los datos para obtener información sobre cómo abordar eventos futuros



Optimization

Proceso que permite encontrar alternativas de rendimiento más rentables o alcanzables, bajo algunas restricciones: maximizando los factores deseados y minimizando los no deseados



¿SUPERVISADO?
¿NO SUPERVISADO?
...

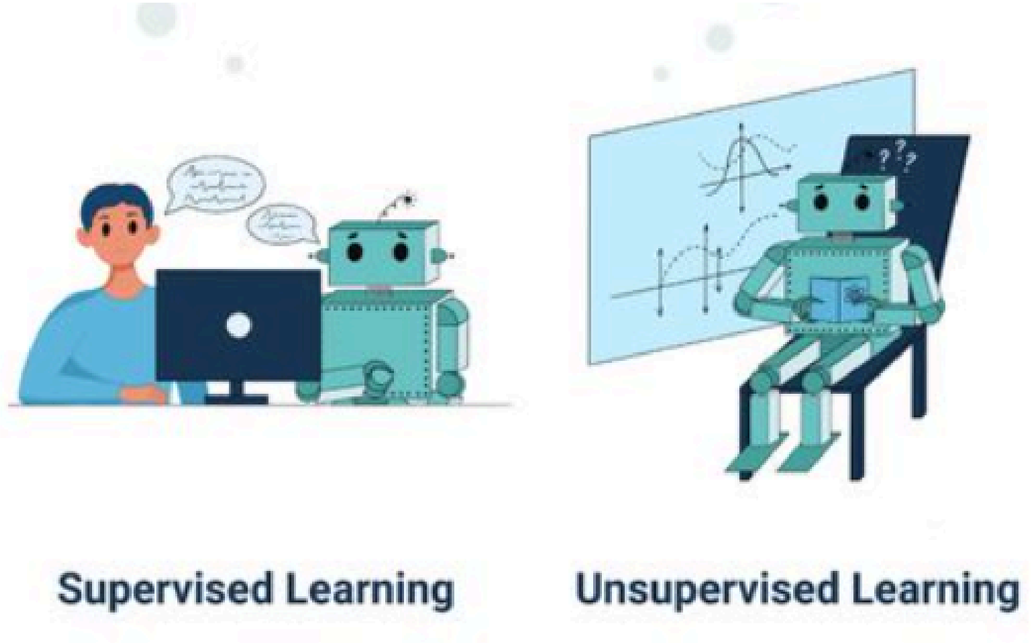


Imagen: https://blog.autify.com/machine-learning-in-software-testing

CLASSICAL MACHINE LEARNING

Data is pre-categorized or numerical

SUPERVISED

Predict a category

CLASSIFICATION

«Divide the socks by color»



Predict a number

REGRESSION

«Divide the ties by length»



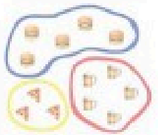
Data is not labeled in any way

UNSUPERVISED

Divide by similarity

CLUSTERING

«Split up similar clothing into stacks»



Find hidden dependencies

DIMENSION REDUCTION (generalization)

«Make the best outfits from the given clothes»



Identify sequences

ASSOCIATION

«Find what clothes I often wear together»



Imagen: https://medium.com/@FhelDimaano/what-is-machine-learning-885aa35db58b

¿SUPERVISADO?
¿NO SUPERVISADO?
¿REINFORCEMENT LEARNING?

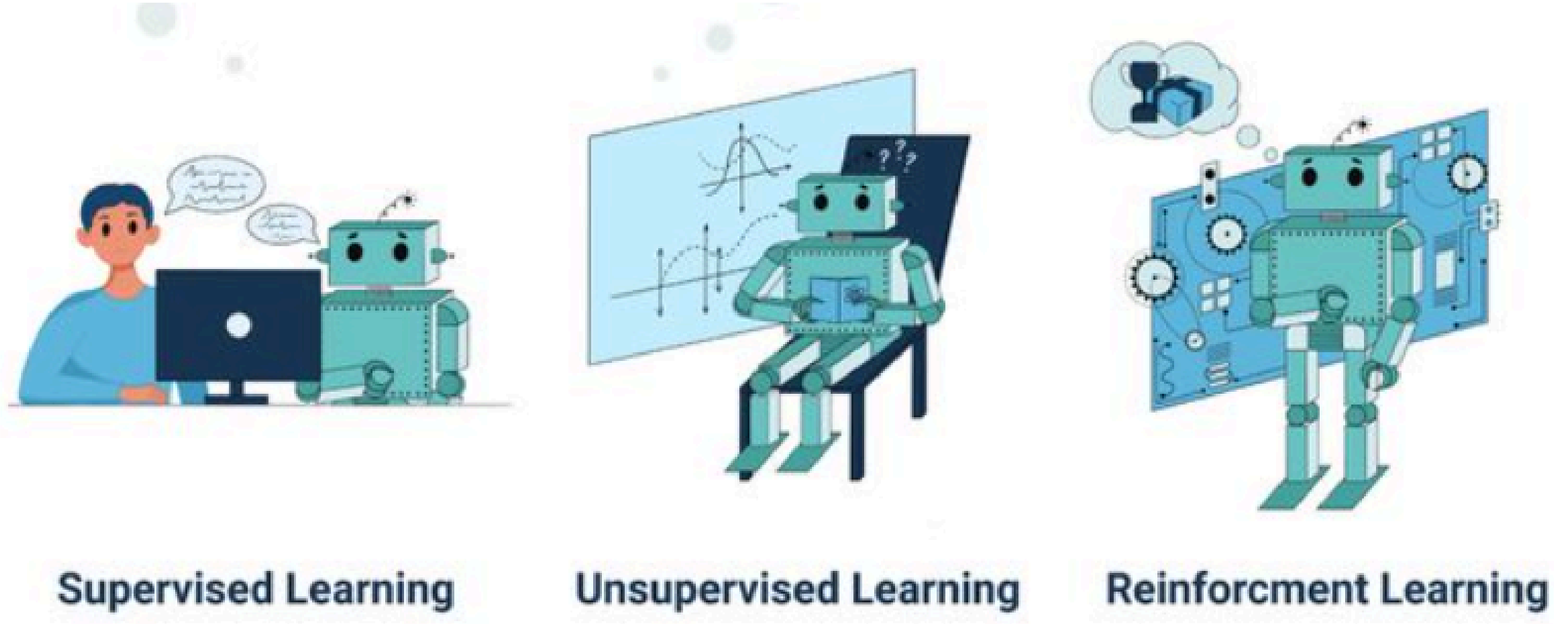


Imagen: https://blog.autify.com/machine-learning-in-software-testing

… MORE ?

Types of Learning

Learning Problems

- 1. Supervised Learning
- 2. Unsupervised Learning
- 3. Reinforcement Learning

Hybrid Learning Problems

- 4. Semi-Supervised Learning
- 5. Self-Supervised Learning
- 6. Multi-Instance Learning

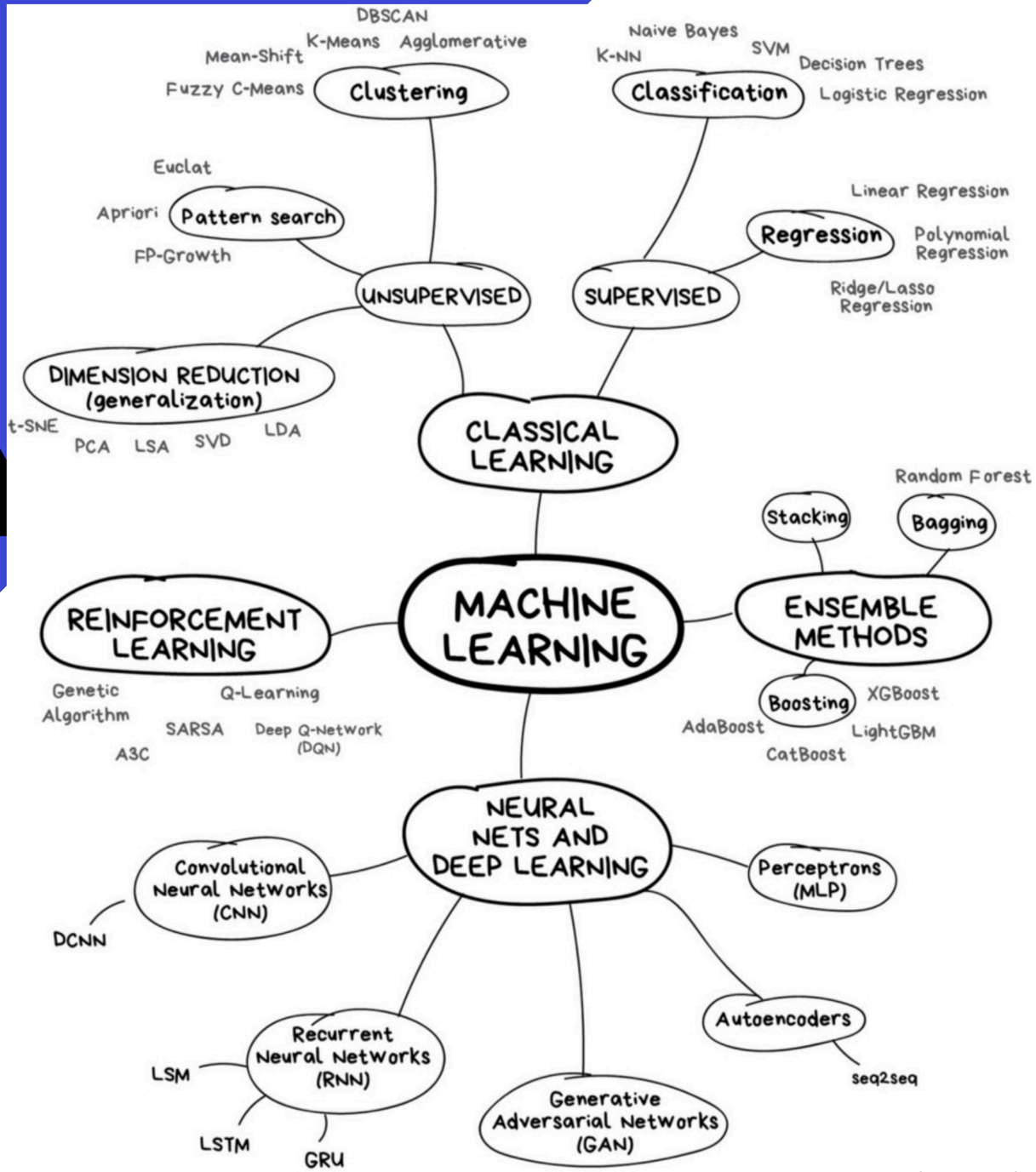
Statistical Inference

- 7. Inductive Learning
- 8. Deductive Inference
- 9. Transductive Learning

Learning Techniques

- 10. Multi-Task Learning
- 11. Active Learning
- 12. Online Learning
- 13. Transfer Learning
- 14. Ensemble Learning

Fuente: https://machinelearningmastery.com/types-of-learning-in-machine-learning/



WHICH TECHNIQUE SHOULD I CHOOSE?

Imagen: https://medium.com/@FhelDimaano/what-is-machine-learning-885aa35db58b

Métodos más utilizados según objetivo

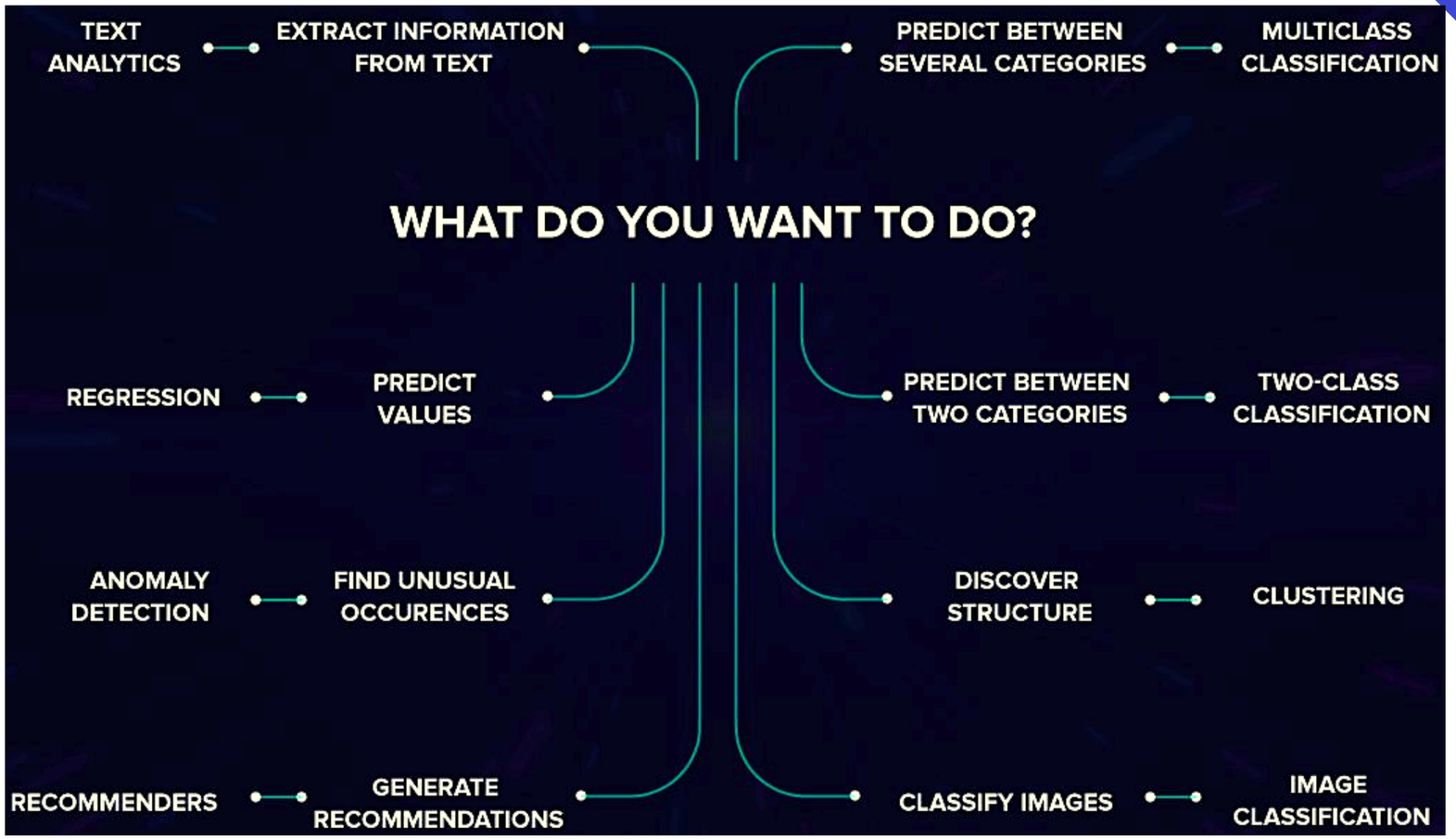


Imagen: https://serokell.io/blog/how-to-choose-ml-technique

Métodos más utilizados según tipo de variable

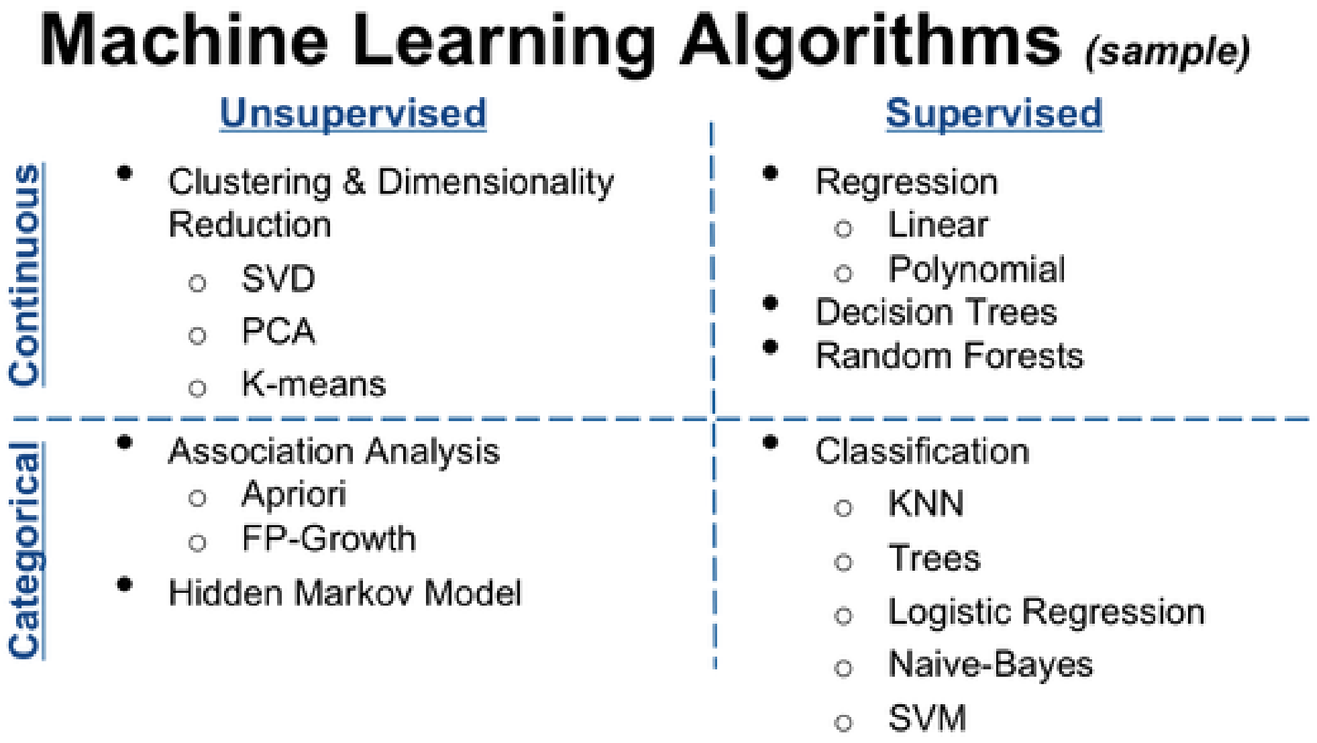


Imagen: http://aiitalia.it/2017/10/23/clustering-machine-learning/

Métodos más utilizados según INPUT

By input:

- If you have a set of labeled data or can prepare such a set, it is the domain of supervised learning.
- If you still need to define a structure, it's an unsupervised learning problem.
- If you need the model to interact with an environment, you will apply a reinforcement learning algorithm.

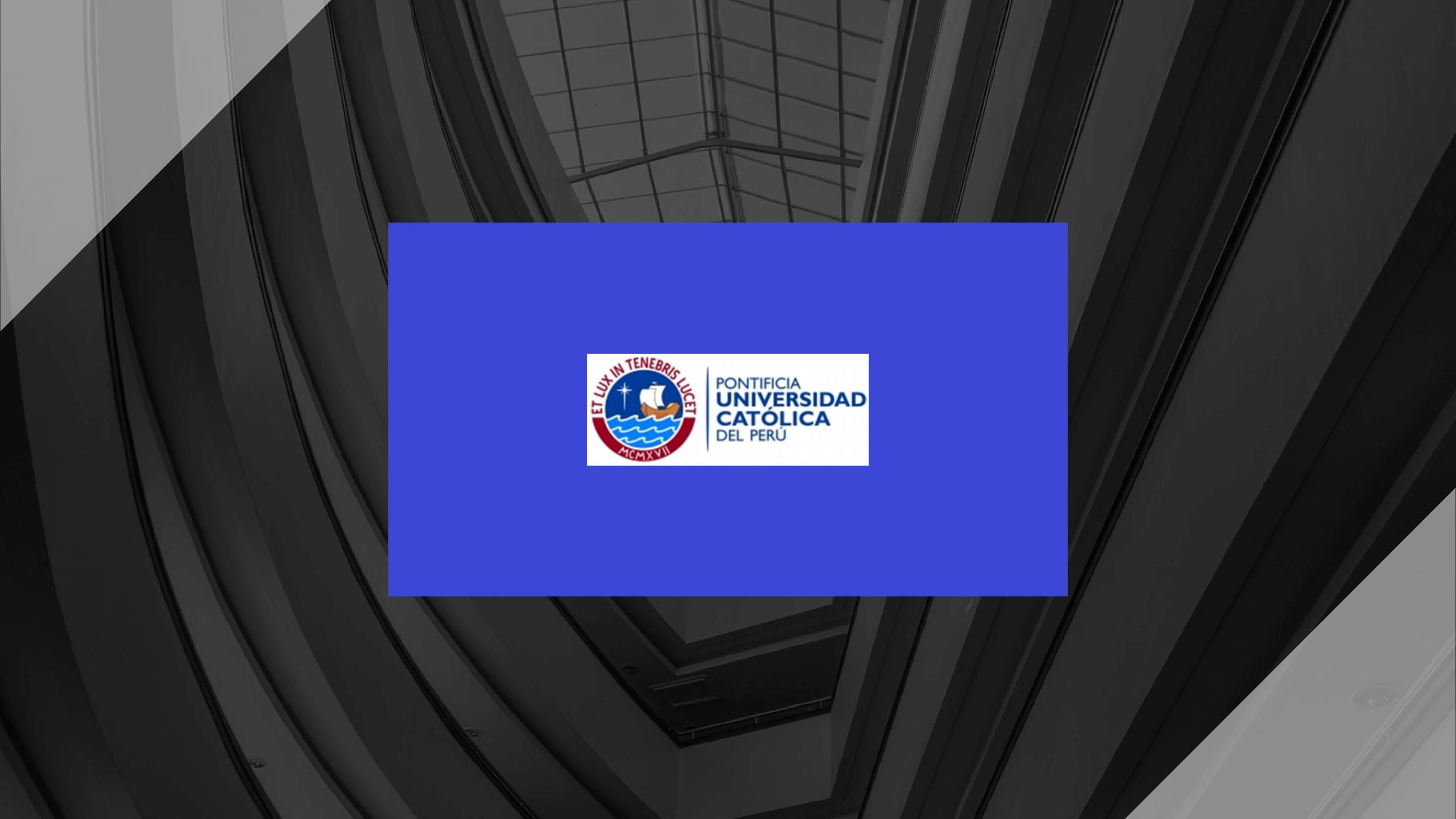
Imagen: https://serokell.io/blog/how-to-choose-ml-technique

Métodos más utilizados según OUTPUT

By output:

- If the output of the model is a number, it's a regression problem.
- If the output of the model is a class and the number of expected classes is known, it's a classification problem.
- If the output of the model is a class but the number of expected classes is unknown, it's a clustering problem.
- If you need to improve performance, it's optimization.
- If you want a system to offer options based on the history of actions, it's a recommendation problem.
- If you want to obtain insights from data, apply pattern recognition models.
- If you want to detect problems, use anomaly detection algorithms.

Imagen: https://serokell.io/blog/how-to-choose-ml-technique


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